

LaTeX

PENG

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1 Introduction

<https://latex-project.org/>

2 TikZ

2.1 Introduction

<https://tikz.dev/>

Definition 2.1 (TikZ). *TikZ is a frontend of PGF.*

2.2 Curved Path Construction

Definition 2.2 (control points). `<ps> .. controls <pc1> and <pc2> .. <pe>`

```
\begin{tikzpicture}[scale=2]
  \filldraw [gray]
    (0,0) circle (2pt)
    (1,1) circle (2pt)
    (2,1) circle (2pt)
    (2,0) circle (2pt);
  \draw [thick] (0,0) .. controls (1,1) and (2,1) .. (2,0);
  \draw [-latex] (0,0) -- node[sloped,above] {tangent} (1,1);
  \draw [-latex] (2,0) -- node[sloped,below] {tangent} (2,1);
\end{tikzpicture}
```

